

Data Converters

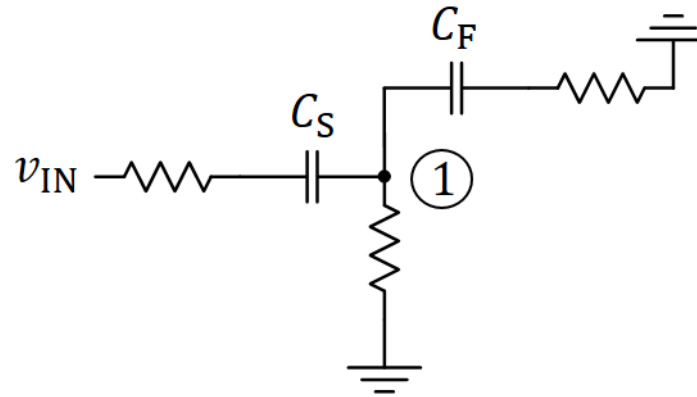
Exercise V

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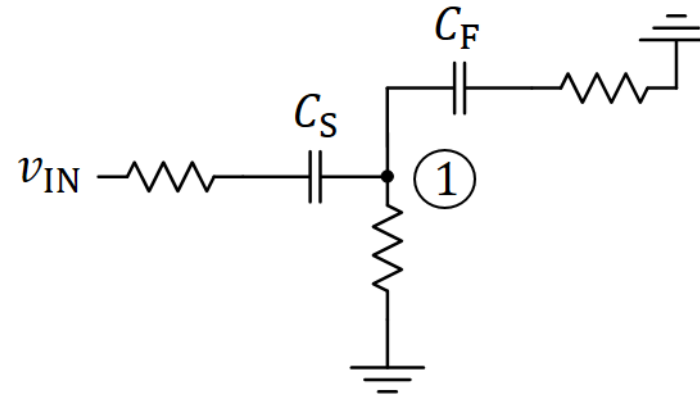
TI.a – Effective Capacitor C_{eff}



According to the paper *Thermal Noise in Track-and-Hold Circuits* by B. Murmann, the effective capacitor C_{eff} results in

$$C_{\text{eff}} = C_S + C_F$$

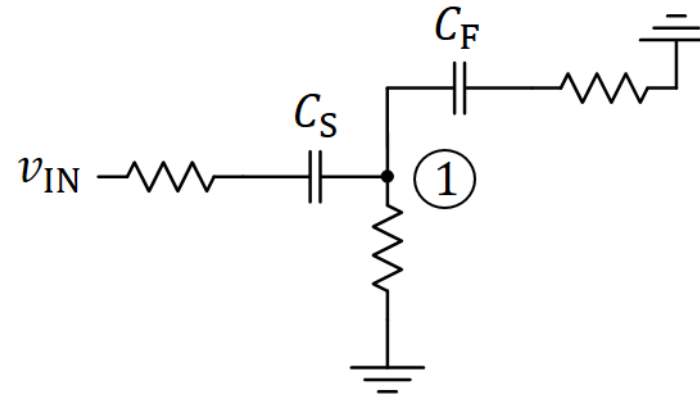
TI.b – Average Squared Charge



Applying the result of task I.a, i.e., $C_{\text{eff}} = C_S + C_F$, to the equipartition theorem yields

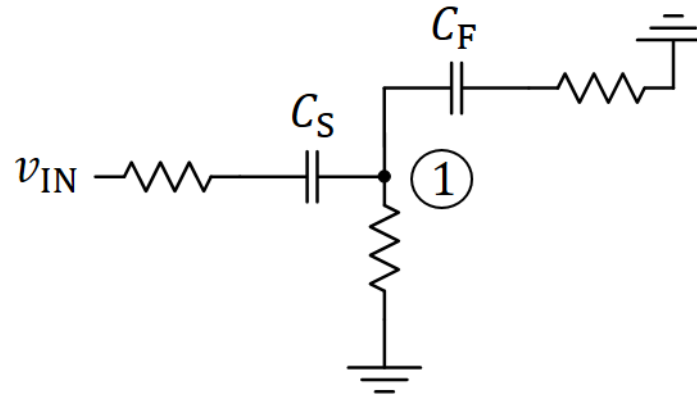
$$\frac{1}{2} \frac{\overline{q_1^2}}{C_{\text{eff}}} = \frac{1}{2} kT \rightarrow \overline{q_1^2} = kT(C_S + C_F)$$

TI.c – Transfer Function (Redistribution Phase)



Likewise the sampled signal, i.e., the charge sampled on C_S ,
**the average noise charge will be transferred by $1/C_F$ to the output
in the redistribution phase.**

TI.d – Total Output Noise of the Sampling Phase

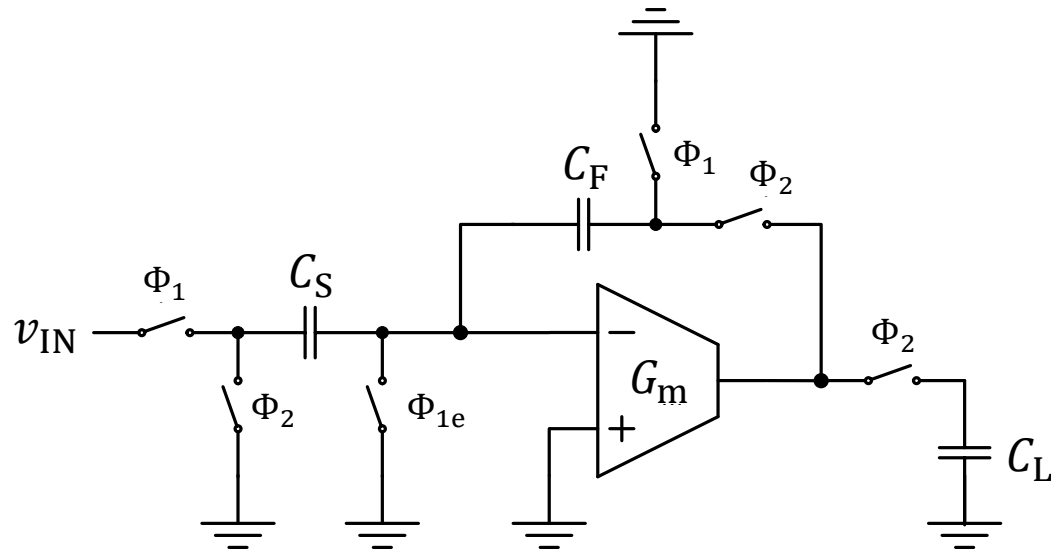


Likewise the sampled signal, i.e., the charge sampled on C_S , the average noise charge will be transferred by $1/C_F$ to the output in the redistribution phase.

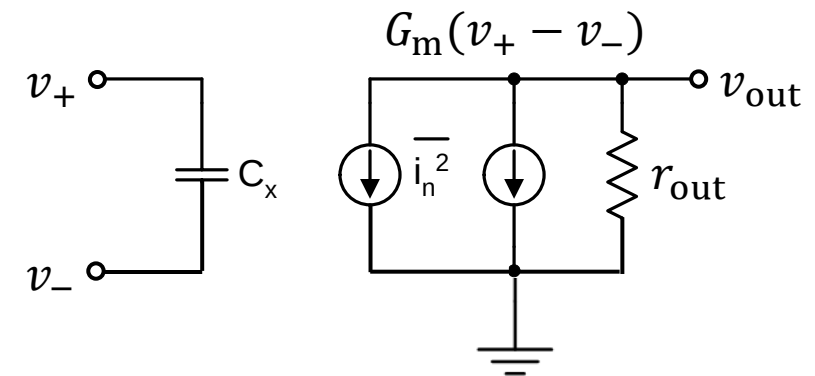
Thus, it will thus be converted into a noise voltage PSD according to

$$\overline{v_{n,sp}^2} = \frac{\overline{q_1^2}}{C_F^2} = \frac{kT(C_S + C_F)}{C_F^2} = \left(1 + \frac{C_S}{C_F}\right) \frac{kT}{C_F}$$

TII.a – Voltage PSD of G_m -cell



charge redistribution track & hold circuit



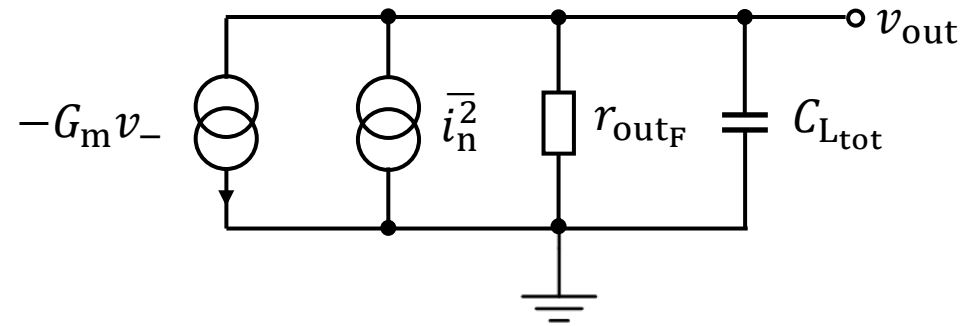
modell of the G_m -cell with

$$\overline{i_n^2} = n_f \cdot 4kT\gamma G_m$$

used for the analysis

TII.a – Voltage PSD of G_m -cell

In PE IV, the overall output of the feedback system was derived and resulted in

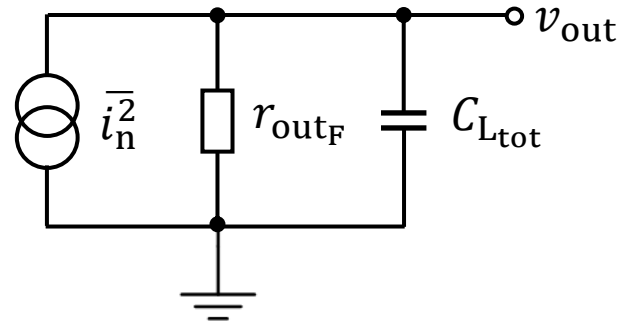


with $r_{outF} = \frac{1}{\beta G_m}$, $C_{Ltot} = C_L + (1 - \beta)C_F$ and $\overline{i_n^2} = n_f \cdot 4kT\gamma G_m$

where $\beta = \frac{C_F}{C_F + C_S + C_X}$

TII.a – Voltage PSD of G_m -cell

For the noise analysis of the G_m -cell, the model simplifies to

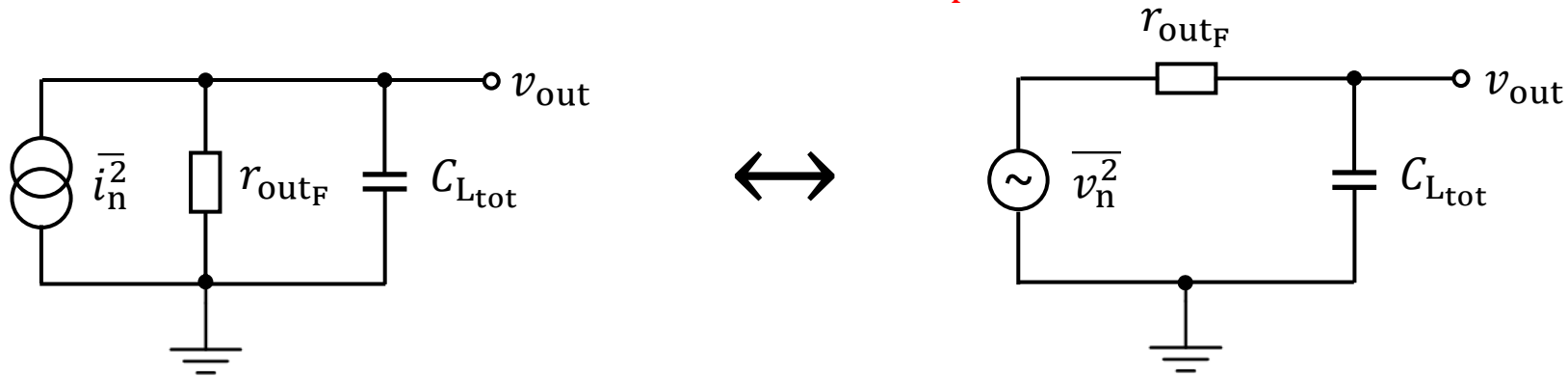


with $r_{outF} = \frac{1}{\beta G_m}$, $C_{Ltot} = C_L + (1 - \beta)C_F$ and $\overline{i_n^2} = n_f \cdot 4kT\gamma G_m$

where $\beta = \frac{C_F}{C_F + C_S + C_X}$

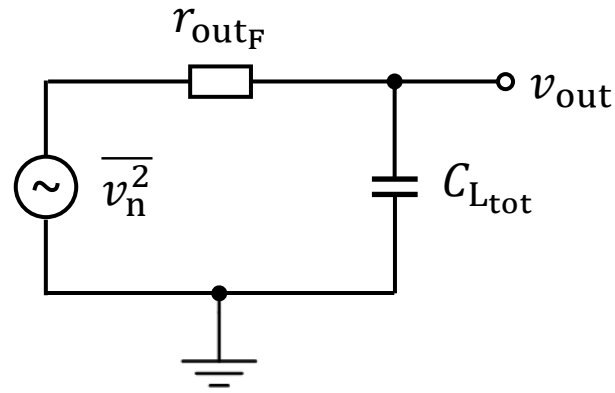
TII.a – Voltage PSD of G_m -cell

This current noise model can easily be converted into the voltage noise model by means of r_{outF} .



$$\overline{v_n^2} = r_{outF}^2 \overline{i_n^2} = \frac{1}{\beta^2 G_m^2} \cdot n_f \cdot 4kT\gamma G_m = n_f \cdot \frac{4kT\gamma}{\beta^2 G_m}$$

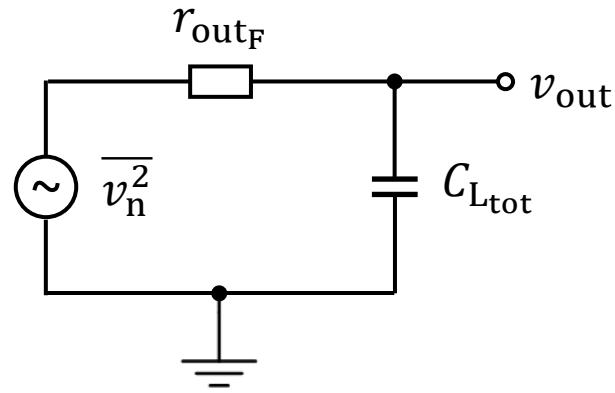
TII.b – Effective Noise-Bandwidth



As discussed in *Mixed-signal CMOS Circuit Design*, the noise bandwidth results in

$$NBW = \frac{p}{4} = \frac{\pi}{2} f = \frac{1}{4r_{outF}C_{Ltot}} = \frac{1}{4} \frac{\beta G_m}{C_{Ltot}}$$

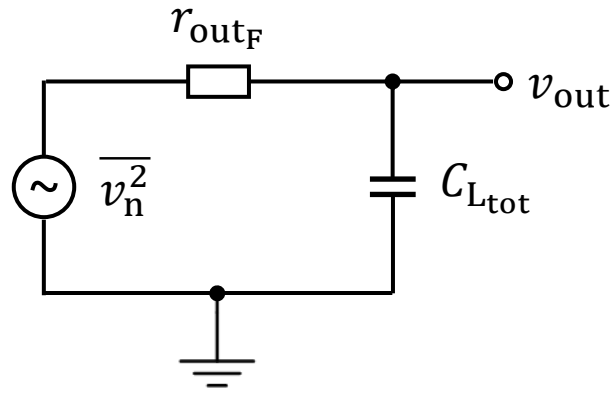
TII.c – Noise Power of G_m -cell



With the knowledge of the NBW, the noise contribution of the G_m -cell to the overall output noise can easily be calculated and results in

$$\overline{v_{n,out}^2} = NBW \cdot \overline{v_n^2} = \frac{1}{4} \frac{\beta G_m}{C_{Ltot}} n_f \frac{4kT\gamma}{\beta^2 G_m} = \frac{n_f \gamma}{\beta} \frac{kT}{C_{Ltot}}$$

TII.d – Total Output-Referred Noise



The total output-referred noise $\overline{v_{n,out,total}^2}$ equals the noise of the sampling phase (since it is transferred to the output in the redistribution phase) plus the noise of the G_m -cell (from the redistribution phase).

$$\overline{v_{n,out,total}^2} = \overline{v_{n,sp}^2} + \overline{v_{n,out}^2} = \left(1 + \frac{C_S}{C_F}\right) \frac{kT}{C_F} + \frac{n_f \gamma}{\beta} \frac{kT}{C_{Ltot}}$$

Discussion – Dominance Noise G_m -Cell

Comparison power spectral densities (PSDs)

$$PSD_{R_{on,S}} = 4kT R_{on,S}$$

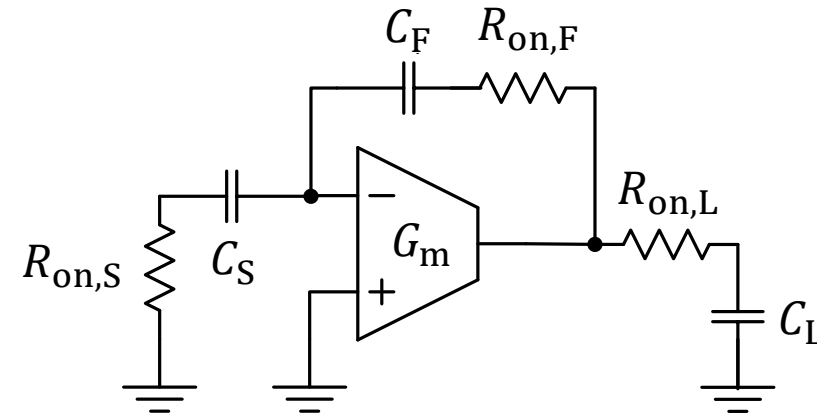
$$PSD_{R_{on,F}} = 4kT R_{on,F}$$

$$PSD_{R_{on,L}} = 4kT R_{on,L}$$

$$PSD_{G_m} = 4kT \frac{n_f \gamma}{\beta^2 G_m} \geq 4kT \frac{\gamma}{G_m}$$

γ/G_m should be (much) larger than any R_{on} !

This requirement is met in any power-efficient design since the settling behavior should be determined by the G_m -cell (and not by the switches).



Discussion – Dominance Noise G_m -Cell

In your argumentation, you only considered the power spectral densities.
However, which impact has the respective transfer function of a noise source on the overall noise power at the output of the circuit?

While the numerators and thus the zeros of the transfer functions may vary, they all have the same denominator (= characteristic equation of the system) and thus the same pole.

Neglecting the zero(es) of every transfer functions since typically $z_{\text{dominant}} > p_{\text{dominant}}$, every noise power is calculated using the same equivalent noise bandwidth.

Thus, all PSDs will just be multiplied by the same factor, i.e., the effective noise bandwidth, which has no impact on the dominance.

T III – Numerical Example

relevant parameters: $G_m = 2 \text{ mS}$, $g_{\text{out}} = 2 \text{ } \mu\text{S}$, $C_S = C_F = 0.1 \text{ pF}$, $C_X = 0.021 \text{ pF}$, $C_L = 0.5 \text{ pF}$

note: $g_{\text{out}} = 2 \text{ } \mu\text{S} \ll G_m = 2 \text{ mS}$

$$TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{-C_S(G_m - sC_F)}{(C_F + C_S + C_X)g_{\text{out}} + C_F G_m + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)}$$
$$\approx \frac{-C_S(G_m - sC_F)}{C_F G_m + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)}$$

T III.a – Small-signal Parameters

relevant parameters: $G_m = 2 \text{ mS}$, $g_{\text{out}} = 2 \text{ } \mu\text{S}$, $C_S = C_F = 0.1 \text{ pF}$, $C_X = 0.021 \text{ pF}$, $C_L = 0.5 \text{ pF}$

$$TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} \approx \frac{-C_S(G_m - sC_F)}{C_F G_m + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)}$$

DC-gain: $A_{\text{DC}} = TF(s = 0) = -1$

feedback factor: $\beta = \frac{C_F}{C_F + C_S + C_X} \approx 0.45$

zero: $f_z = \frac{G_m}{2\pi C_F} \approx 3.18 \text{ GHz}$

pole: $f_p = \frac{\beta G_m}{2\pi[C_L + (1 - \beta)C_F]} \approx 0.258 \text{ GHz}$

note: $f_z = 3.18 \text{ GHz} > 10 \times f_p = 10 \times 0.258 \text{ GHz}$

T III.b – Static Settling Error

relevant parameters: $\beta \approx 0.45, 20 \cdot \log_{10} G_m r_{\text{out}} = 60 \text{ dB}$

$$\varepsilon_s = \frac{TF(s) - TF_{\text{ideal}}(s)}{TF_{\text{ideal}}(s)} = \frac{\frac{A_{L\infty}(s)L(s)}{1 + L(s)} - A_{L\infty}(s)}{A_{L\infty}(s)} = -\frac{1}{1 + L(s)} \approx -\frac{1}{L(s)} = -\frac{1}{-\beta G_m r_{\text{out}}}$$

$$\varepsilon_s \approx \mathbf{0.22\%}$$

T III.c – Dynamic Settling Error (w/o Slewing)

relevant parameters: $G_m = 2 \text{ mS}$, $\frac{g_m}{I_D} = \frac{G_m}{I_D} = 15 \frac{\text{S}}{\text{A}}$, $v_{id} = 10 \text{ mV}$

maximum current: $I_{\max} = 2 \frac{G_m}{G_m/I_D} = 2 \frac{2 \text{ mS}}{15 \text{ S/A}} \approx 0.266 \text{ mA}$

required current: $i = G_m v_{id} = 2 \text{ mS} \cdot 10 \text{ mV} = 0.02 \text{ mA}$

Slewing is not an issue!

T III.c – Dynamic Settling Error (w/o Slewing)

relevant parameters: $\varepsilon_d \approx 0.2\%$, $G_m = 2 \text{ mS}$, $\beta \approx 0.45$, $C_F = 0.1 \text{ pF}$, $C_L = 0.5 \text{ pF}$

$$\varepsilon_d = \frac{TF(t) - TF_{\text{ideal}}(s)}{TF_{\text{ideal}}(s)} = \frac{-\frac{C_S}{C_F}(1 - e^{-t \cdot p})v_{\text{step}} - \left(-\frac{C_S}{C_F}\right)v_{\text{step}}}{-\frac{C_S}{C_F}v_{\text{step}}} = -e^{-t/\tau}$$

$$\rightarrow t = -\ln(-\varepsilon_d) \cdot \tau = -\ln(-\varepsilon_d) \cdot \frac{1}{p} = -\ln(-\varepsilon_d) \frac{C_L + (1 - \beta)C_F}{\beta G_m} \approx 3.8 \text{ nS}$$

T III.d – Dynamic Settling Error (w slewing)

relevant parameters: $G_m = 2 \text{ mS}$, $\frac{g_m}{I_D} = \frac{G_m}{I_D} = 15 \frac{\text{S}}{\text{A}}$, $v_{id} = 800 \text{ mV}$

maximum current: $I_{\max} = 2 \frac{G_m}{G_m/I_D} = 2 \frac{2 \text{ mS}}{15 \text{ S/A}} \approx 0.266 \text{ mA}$

required current: $i = G_m v_{id} = 2 \text{ mS} \cdot 800 \text{ mV} = 1.6 \text{ mA}$

Slewing is an issue!

T III.d – Dynamic Settling Error (Slewing)

$$t > \Delta t_{\text{slew}}: \varepsilon_d = \frac{v_{\text{out}}(t) - V_{\text{OUT}}}{V_{\text{OUT}}} = \frac{e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \Delta v_{\text{settle}}}{V_{\text{OUT}}}$$

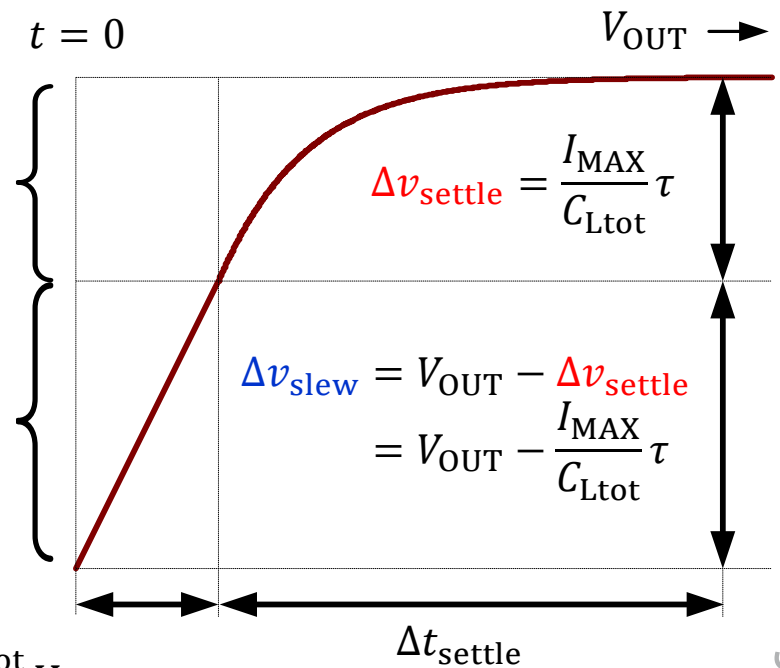
$$V_{\text{OUT}} = -(1 - \varepsilon_s) \frac{C_S}{C_F} v_{\text{step}}$$

$$\tau \approx \frac{C_L + (1 - \beta)C_F}{\beta G_m} = \frac{C_{\text{Ltot}}}{\beta G_m}$$

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$

$$\Delta t_{\text{slew}} = \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} V_{\text{OUT}} - \tau$$



T III.d – Dynamic Settling Error (Slewing)

$$t > \Delta t_{\text{slew}}: \varepsilon_d = \frac{v_{\text{out}}(t) - V_{\text{OUT}}}{V_{\text{OUT}}} = \frac{e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \Delta v_{\text{settle}}}{V_{\text{OUT}}}$$

In order to calculate the settling time t for a dynamic settling error $\varepsilon_d = -0.2 \%$, we have to find V_{OUT} , τ , Δt_{slew} , and Δv_{settle} .

$$V_{\text{OUT}} = -(1 - \varepsilon_s) \frac{C_S}{C_F} v_{\text{step}} = -(1 - 0.22 \%) \cdot 0.8\text{V} = -798.2 \text{ mV}$$

$$\tau \approx \frac{C_{\text{Ltot}}}{\beta G_m} = \frac{C_L + (1 - \beta)C_F}{\beta G_m} = \frac{0.5 \text{ pF} + (1 - 0.45) \cdot 0.1 \text{ pF}}{0.45 \cdot 2 \text{ mS}} \approx 0.62 \text{ ns}$$

T III.d – Dynamic Settling Error (Slewing)

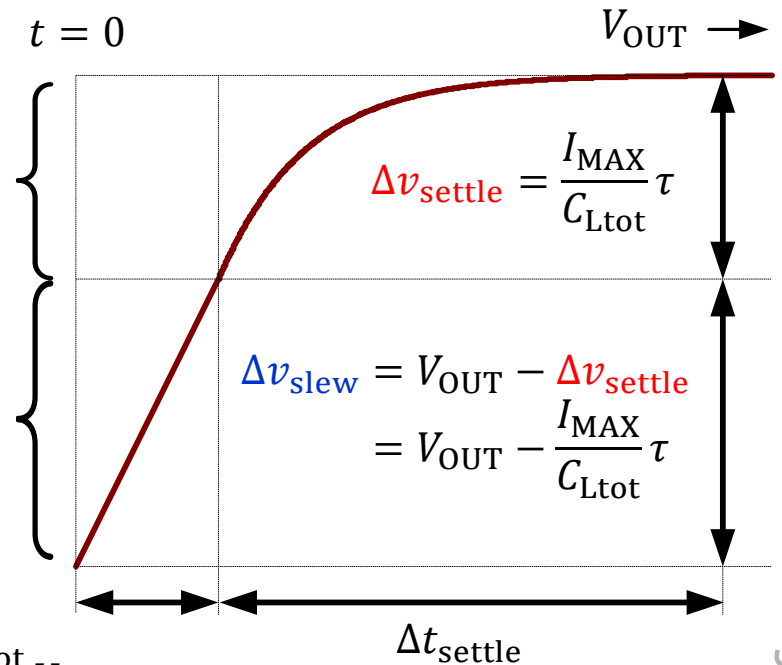
$$t > \Delta t_{\text{slew}}: \varepsilon_d = \frac{v_{\text{out}}(t) - V_{\text{OUT}}}{V_{\text{OUT}}} = \frac{e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \Delta v_{\text{settle}}}{V_{\text{OUT}}}$$

$$\begin{aligned} \Delta v_{\text{settle}} &= -\frac{I_{\text{MAX}}}{C_{\text{Ltot}}} \tau \\ &= \frac{0.266 \text{ mA}}{555 \text{ fF}} 0.62 \text{ ns} \approx -297 \text{ mV} \end{aligned}$$

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$

$$\Delta t_{\text{slew}} = \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} V_{\text{OUT}} - \tau$$



T III.d – Dynamic Settling Error (Slewing)

$$t > \Delta t_{\text{slew}}: \varepsilon_d = \frac{v_{\text{out}}(t) - V_{\text{OUT}}}{V_{\text{OUT}}} = \frac{e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \Delta v_{\text{settle}}}{V_{\text{OUT}}}$$

$$\Delta t_{\text{slew}} = \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} V_{\text{OUT}} - \tau$$

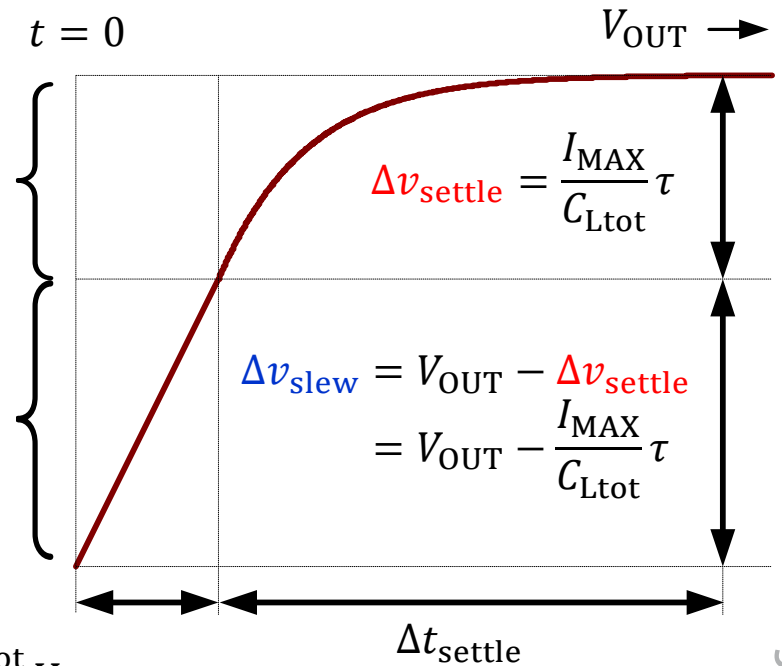
$$= \frac{555 \text{ fF}}{0.266 \text{ mA}} 798.2 \text{ mV} - 0.62 \text{ ns}$$

$$\approx 1.045 \text{ ns}$$

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$

$$\Delta t_{\text{slew}} = \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} V_{\text{OUT}} - \tau$$



T III.d – Dynamic Settling Error (Slewing)

$$t > \Delta t_{\text{slew}}: \varepsilon_d = \frac{v_{\text{out}}(t) - V_{\text{OUT}}}{V_{\text{OUT}}} = \frac{e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \Delta v_{\text{settle}}}{V_{\text{OUT}}}$$

$$V_{\text{OUT}} = -798.2 \text{ mV} \quad \tau \approx 0.62 \text{ ns} \quad \Delta v_{\text{settle}} \approx -297 \text{ mV} \quad \Delta t_{\text{slew}} \approx 1.045 \text{ ns}$$

$$t = -\tau \cdot \ln\left(\frac{\varepsilon_d \cdot V_{\text{OUT}}}{\Delta v_{\text{settle}}}\right) + \Delta t_{\text{slew}} \approx 4.29 \text{ ns}$$

The total output-referred noise $\overline{v_{n,\text{out},\text{total}}^2}$ equals the noise of the sampling phase (since it is transferred to the output in the redistribution phase) plus the noise of the G_m -cell (from the redistribution phase).

$$\overline{v_{n,\text{out},\text{total}}^2} = \overline{v_{n,\text{sp}}^2} + \overline{v_{n,\text{out}}^2} = \left(1 + \frac{C_S}{C_F}\right) \frac{kT}{C_F} + \frac{n_f \gamma}{\beta} \frac{kT}{C_{L,\text{tot}}}$$

$$= 1.38 \cdot 10^{-23} \frac{\text{J}}{\text{K}} \cdot 293 \text{ K} \left[\left(1 + \frac{0.1 \text{ pF}}{0.1 \text{ pF}}\right) \frac{1}{0.1 \text{ pF}} + \frac{2 \cdot \frac{2}{3}}{0.45} \frac{1}{0.555 \text{ pF}} \right]$$

$$\approx 80.868 \mu\text{V}^2 + 21.586 \mu\text{V}^2$$

$$v_{n,\text{out},\text{total}}_{\text{RMS}} = \sqrt{\overline{v_{n,\text{out},\text{total}}^2}} \approx \mathbf{320 \mu V}$$